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TREND-BASED MOCK PAPER

GATE Electrical Engineering

Mock Paper — 03

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Electrical Engineering
Paper:	Mock 03
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

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Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

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Section A: General Aptitude

Q1. [1-NAT] [1 mark]

Distance 200km at 50km/h takes:

(Numerical Answer Type — Enter the numerical value)

Q2. [1-NAT] [1 mark]

Choose opposite of 'latent':

(Numerical Answer Type — Enter the numerical value)

Q3. [1-MCQ] [1 mark]

Choose synonym of 'pragmatic':

- (A) Idealistic
- (B) Practical
- (C) Theoretical
- (D) Abstract

Q4. [1-MCQ] [1 mark]

Speed = Distance /:

- (A) Time
- (B) Velocity
- (C) Acceleration
- (D) Force

Section B: Electrical Engineering

Q5. [2-NAT] [2 marks]

Wheatstone bridge null condition:

(Numerical Answer Type — Enter the numerical value)

Q6. [2-MCQ] [2 marks]

Laplace transform of $e^{(-3t)}u(t)$:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) e^{-3}/s

Q7. [2-MCQ] [2 marks]

Y-parameter Y_{11} for series impedance Z is:

- (A) $1/Z$
- (B) Z
- (C) $-Z$
- (D) 0

Q8. [2-MCQ] [2 marks]

Superposition theorem applies to:

- (A) Linear only
- (B) Non-linear
- (C) Both
- (D) Time-varying only

Q9. [2-MCQ] [2 marks]

$H(s) = (s^2 + 4s + 3)/(s^2 + 2s + 2)$. Zeros at:

- (A) $s = -1, -3$
- (B) $s = -1 + j, -1 - j$
- (C) $s = -1, -3$ and $-1 + j$
- (D) $s = -2 + j, -2 - j$

Q10. [2-MSQ] [2 marks]

Norton equivalent of Thevenin ($V_{th} = 10V$, $R_{th} = 5\Omega$):

- (A) $I_n = 2A$, $R_n = 5$
- (B) $I_n = 2A$, $R_n = 10$
- (C) $I_n = 0.5A$, $R_n = 5$
- (D) $I_n = 0.5A$, $R_n = 10$

Q11. [2-NAT] [2 marks]

Two-port h-parameter h_{11} has units:

(Numerical Answer Type — Enter the numerical value)

Q12. [2-MSQ] [2 marks]

Time constant of RL ($R = 10\Omega$, $L = 5H$):

- (A) $2s$
- (B) $0.5s$
- (C) $50s$
- (D) $0.05s$

Q13. [2-NAT] [2 marks]

Delta-Star transformation: R_a (star) =

(Numerical Answer Type — Enter the numerical value)

Q14. [1-MCQ] [1 mark]

Eigenvalues of identity:

- (A) All zero
- (B) All one
- (C) Distinct
- (D) Depends

Q15. [2-NAT] [2 marks]

Which of the following is a probability generating function for Poisson(λ)?

(Numerical Answer Type — Enter the numerical value)

Q16. [2-NAT] [2 marks]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

(Numerical Answer Type — Enter the numerical value)

Q17. [2-MCQ] [2 marks]

If $f(x) = x^3 - 3x + 2$, number of real roots:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q18. [2-NAT] [2 marks]

$P(|Z| < 1)$ for $Z \sim N(0,1)$ is approximately:

(Numerical Answer Type — Enter the numerical value)

Q19. [2-MCQ] [2 marks]

$\frac{d}{dx}[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

- (A) 0
- (B) 2
- (C) 4
- (D) 1

Q20. [2-MCQ] [2 marks]

The vector $(1,2,3)$ projected onto $(2,1,0)$ is:

- (A) $(4/5, 2/5, 0)$
- (B) $(8/5, 4/5, 0)$
- (C) $(2, 1, 0)$
- (D) $(1, 2, 3)$

Q21. [2-MSQ] [2 marks]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

- (A) 1
- (B) 0
- (C) -1
- (D) $\cos(\pi)$

Q22. [1-NAT] [1 mark]

Synchronous generator excitation:

(Numerical Answer Type — Enter the numerical value)

Q23. [1-MCQ] [1 mark]

Circuit breaker operates on:

- (A) Overload
- (B) Short circuit
- (C) Both
- (D) None

Q24. [1-MCQ] [1 mark]

Per unit system base values:

- (A) Voltage only
- (B) Power only
- (C) Voltage and Power
- (D) Current

Q25. [1-MCQ] [1 mark]

Most efficient transmission:

- (A) Low voltage
- (B) High voltage
- (C) DC
- (D) AC

Q26. [2-MSQ] [2 marks]

Swing equation relates:

- (A) Power and voltage
- (B) Accelerating power and rotor angle
- (C) Flux and current
- (D) Torque and speed

Q27. [2-MCQ] [2 marks]

3-phase, 4-pole IM at 1440 rpm on 50 Hz. Slip:

- (A) 2%
- (B) 4%
- (C) 6%
- (D) 8%

Q28. [2-NAT] [2 marks]

100km line: $R=0.1$, $X=0.5$ ohm/km. Short line ABCD:

(Numerical Answer Type — Enter the numerical value)

Q29. [2-MCQ] [2 marks]

Transmission line: sending end voltage higher than receiving for:

- (A) Heavy loading
- (B) Light loading (capacitive)
- (C) Unity PF
- (D) No load

Q30. [2-MCQ] [2 marks]

6600/660V transformer turns ratio (primary:secondary):

- (A) 10:1
- (B) 1:10
- (C) 100:1

(D) 1:100

Q31. [1-MCQ] [1 mark]

Full-wave rectifier uses:

- (A) 1
- (B) 2
- (C) 4
- (D) 3

Q32. [1-MCQ] [1 mark]

Thyristor is:

- (A) Current controlled
- (B) Voltage controlled
- (C) Power
- (D) None

Q33. [1-NAT] [1 mark]

Inverter converts:

(Numerical Answer Type — Enter the numerical value)

Q34. [1-MCQ] [1 mark]

Step-down chopper output:

- (A) Less than input
- (B) More than input
- (C) Equal
- (D) Zero

Q35. [2-MCQ] [2 marks]

Step-down chopper $V_s=200V$, $\alpha=0.6$. Output:

- (A) 120V
- (B) 80V
- (C) 200V
- (D) 333V

Q36. [2-MCQ] [2 marks]

Inductor current in CCM of DC-DC converter:

- (A) Always zero
- (B) Never zero
- (C) Always constant
- (D) Oscillating

Q37. [2-MCQ] [2 marks]

IGBT combines features of:

- (A) BJT and MOSFET
- (B) MOSFET and diode
- (C) BJT and SCR
- (D) MOSFET and JFET

Q38. [2-MCQ] [2 marks]

For a buck converter, $V_{out} =$

- (A) $D \cdot V_{in}$
- (B) V_{in}/D
- (C) $V_{in}/(1-D)$
- (D) $V_{in} \cdot D/(1-D)$

Q39. [1-MCQ] [1 mark]

DC starter used to:

- (A) Regulate speed
- (B) Limit start current
- (C) Reverse
- (D) Increase torque

Q40. [1-MCQ] [1 mark]

Slip in induction motor:

- (A) Always positive
- (B) Can be negative
- (C) Always zero
- (D) Infinity

Q41. [1-MCQ] [1 mark]

Alternator produces:

- (A) DC
- (B) AC
- (C) Both
- (D) None

Q42. [1-MCQ] [1 mark]

Transformer works on:

- (A) Auto induction
- (B) Mutual induction
- (C) Self induction
- (D) None

Q43. [1-MCQ] [1 mark]

Slip $s =$

- (A) $(N_s - N_r)/N_s$
- (B) $(N_r - N_s)/N_s$
- (C) N_s/N_r
- (D) N_r/N_s

Q44. [3-MCQ] [3 marks]

3-phase transformer primary:secondary turns = 10:1. If primary is delta, secondary is:

- (A) Delta
- (B) Star
- (C) Both
- (D) Either

Q45. [2-MCQ] [2 marks]

Over-excited synchronous generator operates at:

- (A) Lagging PF
- (B) Leading PF
- (C) Unity PF
- (D) Zero PF

Q46. [1-MCQ] [1 mark]

Gain margin:

- (A) At phase crossover
- (B) At gain crossover
- (C) Both
- (D) None

Q47. [1-MCQ] [1 mark]

Integral controller eliminates:

- (A) Rise time
- (B) Steady state error
- (C) Damping
- (D) Overshoot

Q48. [1-MCQ] [1 mark]

Time constant of RC:

- (A) RC
- (B) R/C
- (C) C/R
- (D) 1/RC

Q49. [1-NAT] [1 mark]

Root locus plots:

(Numerical Answer Type — Enter the numerical value)

Q50. [2-MCQ] [2 marks]

Root locus branches start at poles and end at:

- (A) Poles only
- (B) Zeros only
- (C) Zeros or infinity
- (D) Infinity only

Q51. [2-MCQ] [2 marks]

Phase margin is defined at:

- (A) Phase crossover
- (B) Gain crossover
- (C) Both
- (D) 0 dB crossing

Q52. [2-MCQ] [2 marks]

Compound interest: P at r% for 2 years =

- (A) $P(1+r/100)^2$
- (B) $P(1+2r/100)$
- (C) $P + Pr/100 \times 2$

(D) $P \cdot r / 100$

Q53. [2-MCQ] [2 marks]

Sum of first n odd numbers:

- (A) n^2
- (B) $2n$
- (C) $n(n+1)$
- (D) $n(n-1)$

Q54. [2-NAT] [2 marks]

If $x + 1/x = 3$, then $x^3 + 1/x^3 =$

(Numerical Answer Type — Enter the numerical value)

Q55. [2-MSQ] [2 marks]

Corona loss increases with:

- (A) Wire diameter
- (B) Voltage
- (C) Both
- (D) None

Q56. [2-MCQ] [2 marks]

Displacement current added by:

- (A) Faraday
- (B) Maxwell
- (C) Ampere
- (D) Gauss

Q57. [1-NAT] [1 mark]

Ideal op-amp gain:

(Numerical Answer Type — Enter the numerical value)

Q58. [1-NAT] [1 mark]

RC phase shift uses:

(Numerical Answer Type — Enter the numerical value)

Q59. [1-MCQ] [1 mark]

Negative feedback:

- (A) Increases gain
- (B) Decreases gain
- (C) No effect
- (D) Oscillates

Q60. [1-NAT] [1 mark]

Universal gate:

(Numerical Answer Type — Enter the numerical value)

Q61. [1-MCQ] [1 mark]

JK flip-flop toggles when:

- (A) $J=K=0$
- (B) $J=K=1$

- (C) $J=0, K=1$
- (D) $J=1, K=0$

Q62. [1-NAT] [1 mark]

Mod-10 counts from:

(Numerical Answer Type — Enter the numerical value)

Q63. [1-MCQ] [1 mark]

LTI characterized by:

- (A) Impulse response
- (B) Step response
- (C) Frequency
- (D) All

Q64. [1-NAT] [1 mark]

Aliasing avoided by:

(Numerical Answer Type — Enter the numerical value)

Q65. [1-NAT] [1 mark]

FS coefficients:

(Numerical Answer Type — Enter the numerical value)

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1NAT	B	1
2	1NAT	A	1
3	1MCQ	B	1
4	1MCQ	A	1
5	1MCQ	B	1
6	1MCQ	B	1
7	1MCQ	A	1
8	1MCQ	B	1
9	1MCQ	B	1
10	1MCQ	A	1
11	1NAT	A	1
12	1MCQ	C	1
13	1MCQ	C	1

14	1MCQ	B	1
15	1MCQ	A	1
16	1MCQ	B	1
17	1MCQ	A	1
18	1NAT	B	1
19	1MCQ	C	1
20	1MCQ	A	1
21	1NAT	A	1
22	1MCQ	A	1
23	1NAT	A	1
24	1NAT	B	1
25	1MCQ	B	1
26	1NAT	C	1
27	1MCQ	B	1
28	1NAT	B	1

29	1MCQ	D	1
30	1NAT	A	1
31	1NAT	A	1
32	2MSQ	C	2
33	2MCQ	B	2
34	2MCQ	A	2
35	2NAT	A	2
36	2MSQ	B	2
37	2MCQ	B	2
38	2MCQ	A	2
39	2MCQ	A	2
40	2NAT	A	2
41	2NAT	C	2
42	2MCQ	A	2
43	2NAT	A	2

44	2MCQ	D	2
45	2MCQ	B	2
46	2MCQ	A	2
47	2MCQ	A	2
48	2NAT	A	2
49	2MCQ	B	2
50	2MCQ	C	2
51	2MSQ	A	2
52	2MCQ	C	2
53	2MCQ	A	2
54	2NAT	A	2
55	2NAT	A	2
56	3MCQ	B	3
57	2MCQ	B	2
58	2MSQ	B	2

59	2MCQ	B	2
60	2MCQ	A	2
61	2NAT	B	2
62	2MCQ	B	2
63	2MCQ	B	2
64	2MSQ	C	2
65	2MCQ	A	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Quantitative | Difficulty: easy

Q2.

Answer: A

Solution:

Correct Answer: A

Subject: General Aptitude | Topic: Verbal | Difficulty: moderate

Q3.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Verbal | Difficulty: moderate

Q4.

Answer: A

Solution:

Correct Answer: A

Subject: General Aptitude | Topic: Quantitative | Difficulty: difficult

Q5.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Linear Algebra | Difficulty: easy

Q6.

Answer: B

Solution:

Correct Answer: B

Subject: Electrical Machines | Topic: DC Machines | Difficulty: easy

Q7.

Answer: A

Solution:

Correct Answer: A

Subject: Electrical Machines | Topic: Induction Motor | Difficulty: easy

Q8.

Answer: B

Solution:

Correct Answer: B

Subject: Electrical Machines | Topic: Alternators | Difficulty: moderate

Q9.

Answer: B

Solution:

Correct Answer: B

Subject: Electrical Machines | Topic: Transformers | Difficulty: difficult

Q10.

Answer: A

Solution:

Correct Answer: A

Subject: Electrical Machines | Topic: Induction Motor | Difficulty: moderate

Q11.

Answer: A

Solution:

Correct Answer: A

Subject: Power Systems | Topic: Generators | Difficulty: easy

Q12.

Answer: C

Solution:

Correct Answer: C

Subject: Power Systems | Topic: Protection | Difficulty: moderate

Q13.

Answer: C

Solution:

Correct Answer: C

Subject: Power Systems | Topic: Per Unit | Difficulty: easy

Q14.

Answer: B

Solution:

Correct Answer: B

Subject: Power Systems | Topic: Transmission | Difficulty: difficult

Q15.

Answer: A

Solution:

Correct Answer: A

Subject: Control Systems | Topic: Frequency Response | Difficulty: moderate

Q16.

Answer: B

Solution:

Correct Answer: B

Subject: Control Systems | Topic: Controllers | Difficulty: moderate

Q17.

Answer: A

Solution:

Correct Answer: A

Subject: Control Systems | Topic: Time Response | Difficulty: moderate

Q18.

Answer: B

Solution:

Correct Answer: B

Subject: Control Systems | Topic: Root Locus | Difficulty: moderate

Q19.

Answer: C

Solution:

Correct Answer: C

Subject: Power Electronics | Topic: Rectifiers | Difficulty: easy

Q20.

Answer: A

Solution:

Correct Answer: A

Subject: Power Electronics | Topic: Converters | Difficulty: moderate

Q21.

Answer: A

Solution:

Correct Answer: A

Subject: Power Electronics | Topic: Inverters | Difficulty: easy

Q22.

Answer: A

Solution:

Correct Answer: A

Subject: Power Electronics | Topic: Choppers | Difficulty: difficult

Q23.

Answer: A

Solution:

Correct Answer: A

Subject: Analog Electronics | Topic: Op-Amps | Difficulty: moderate

Q24.

Answer: B

Solution:

Correct Answer: B

Subject: Analog Electronics | Topic: Oscillators | Difficulty: moderate

Q25.

Answer: B

Solution:

Correct Answer: B

Subject: Analog Electronics | Topic: Feedback | Difficulty: moderate

Q26.

Answer: C

Solution:

Correct Answer: C

Subject: Digital Electronics | Topic: Logic Gates | Difficulty: easy

Q27.

Answer: B

Solution:

Correct Answer: B

Subject: Digital Electronics | Topic: Flip-Flops | Difficulty: moderate

Q28.

Answer: B

Solution:

Correct Answer: B

Subject: Digital Electronics | Topic: Counters | Difficulty: moderate

Q29.

Answer: D

Solution:

Correct Answer: D

Subject: Signals and Systems | Topic: LTI Systems | Difficulty: easy

Q30.

Answer: A

Solution:

Correct Answer: A

Subject: Signals and Systems | Topic: Sampling | Difficulty: moderate

Q31.

Answer: A

Solution:

Correct Answer: A

Subject: Signals and Systems | Topic: Fourier | Difficulty: moderate

Q32.

Answer: C

Solution:

Correct Answer: C

Subject: EMFT | Topic: Transmission | Difficulty: moderate

Q33.

Answer: B

Solution:

Correct Answer: B

Subject: EMFT | Topic: Maxwell | Difficulty: moderate

Q34.

Answer: A

Solution:

$CI = P(1+r/100)^n$. For 2 years: $P(1+r/100)^2$.

Subject: Measurements | Topic: Measurements | Difficulty: moderate

Q35.

Answer: A

Solution:

PGF of Poisson(λ): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q36.

Answer: B

Solution:

$M^2 d^2(\delta)/dt^2 = P_m - P_e$. Relates accelerating power to rotor angle dynamics.

Subject: Power Systems | Topic: Power Systems | Difficulty: easy

Q37.

Answer: B

Solution:

$N_{sync} = 120 \cdot 50 / 4 = 1500$. $s = (1500 - 1440) / 1500 = 60 / 1500 = 0.04 = 4\%$.

Subject: Power Systems | Topic: Power Systems | Difficulty: moderate

Q38.

Answer: A

Solution:

$1 + 3 + 5 + \dots + (2n-1) = n^2$.

Subject: Measurements | Topic: Measurements | Difficulty: moderate

Q39.

Answer: A

Solution:

$V_o = \alpha \cdot V_s = 0.6 \cdot 200 = 120V$.

Subject: Power Electronics | Topic: Power Electronics | Difficulty: moderate

Q40.

Answer: A

Solution:

Short line: only series Z. $Z = (0.1 + j0.5) \cdot 100 = 60 \text{ ohm}$. $A=1$, $B=60$, $C=0$, $D=1$.

Subject: Power Systems | Topic: Power Systems | Difficulty: difficult

Q41.

Answer: C

Solution:

Null: $R_1/R_2 = R_3/R_4$, which implies $R_1 \cdot R_4 = R_2 \cdot R_3$.

Subject: Network Theory | Topic: Network Theory | Difficulty: moderate

Q42.

Answer: A

Solution:

$L\{e^{-(s+a)}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Network Theory | Topic: Network Theory | Difficulty: easy

Q43.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4} \approx 0.908$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q44.

Answer: D

Solution:

Series Z: $I_1=I_2$, $V_1-V_2=Z \cdot I_1$. $Y_{11} = I_1/V_1$ with $I_2=0$ impossible ($I_1=I_2$). $Y_{11}=0$ for pure series.

Subject: Network Theory | Topic: Network Theory | Difficulty: moderate

Q45.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q46.

Answer: A

Solution:

Superposition requires linearity (homogeneity + additivity). Does NOT apply to non-linear.

Subject: Network Theory | Topic: Network Theory | Difficulty: easy

Q47.

Answer: A

Solution:

Zeros from numerator: $s^2+4s+3 = (s+1)(s+3)$. Zeros at $s=-1, -3$.

Subject: Network Theory | Topic: Network Theory | Difficulty: easy

Q48.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q49.

Answer: B

Solution:

Ferranti effect: light load/capacitive charging causes $V_s > V_r$. Long lines at no-load.

Subject: Power Systems | Topic: Power Systems | Difficulty: moderate

Q50.

Answer: C

Solution:

Root locus: starts at OL poles, ends at OL zeros (or infinity if $P > Z$).

Subject: Control Systems | Topic: Control Systems | Difficulty: difficult

Q51.

Answer: A

Solution:

$I_n = V_{th}/R_{th} = 10/5 = 2A$. $R_n = R_{th} = 5\Omega$.

Subject: Network Theory | Topic: Network Theory | Difficulty: moderate

Q52.

Answer: C

Solution:

$d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $d/dx \ln(\sec x) = \tan x$. $d/dx \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = d/dx[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q53.

Answer: A

Solution:

$a = V_1/V_2 = 6600/660 = 10$. Primary:secondary = 10:1.

Subject: Power Systems | Topic: Power Systems | Difficulty: difficult

Q54.

Answer: A

Solution:

$h_{11} = (V_1/I_1)$ with output shorted. Units: $V/A = \Omega$.

Subject: Network Theory | Topic: Network Theory | Difficulty: moderate

Q55.

Answer: A

Solution:

Cube: $(x+1/x)^3 = x^3 + 3x + 3/x + 1/x^3 = 27$. $x^3 + 1/x^3 = 27 - 3 = 24$.

Subject: Measurements | Topic: Measurements | Difficulty: moderate

Q56.

Answer: B

Solution:

Dy connection: primary delta, secondary star. Common for step-down (reduces voltage, provides neutral).

Subject: Electrical Machines | Topic: Electrical Machines | Difficulty: moderate

Q57.

Answer: B

Solution:

Continuous conduction mode (CCM): inductor current never reaches zero.

Subject: Power Electronics | Topic: Power Electronics | Difficulty: easy

Q58.

Answer: B

Solution:

$\tau = L/R = 5/10 = 0.5$ s.

Subject: Network Theory | Topic: Network Theory | Difficulty: moderate

Q59.

Answer: B

Solution:

Phase margin: at gain crossover ($|GH| = 1$), $1 + \text{phase in degrees}$.

Subject: Control Systems | Topic: Control Systems | Difficulty: moderate

Q60.

Answer: A

Solution:

IGBT: BJT (high current) + MOSFET (easy drive). High input impedance, low saturation voltage.

Subject: Power Electronics | Topic: Power Electronics | Difficulty: easy

Q61.

Answer: B

Solution:

$R_a = (R_1 R_2 + R_2 R_3 + R_3 R_1) / R_3$.

Subject: Network Theory | Topic: Network Theory | Difficulty: moderate

Q62.

Answer: B

Solution:

$\text{proj} = ((v \cdot w) / (w \cdot w)) * w = (4/5) * (2, 1, 0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q63.

Answer: B

Solution:

Over-excitation: generator supplies reactive power, operates at leading power factor.

Subject: Electrical Machines | Topic: Electrical Machines | Difficulty: moderate

Q64.

Answer: C

Solution:

$y = A \cos(2x) + B \sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q65.

Answer: A

Solution:

Buck (step-down): $V_o = D * V_{in}$ where D is duty cycle.

Subject: Power Electronics | Topic: Power Electronics | Difficulty: easy

PAPER BLUEPRINT

Subject Distribution:

Network Theory: 9 questions, 18 marks
Engineering Mathematics: 8 questions, 15 marks
Power Systems: 9 questions, 14 marks
Power Electronics: 8 questions, 12 marks
Electrical Machines: 7 questions, 10 marks
Control Systems: 6 questions, 8 marks
Measurements: 3 questions, 6 marks
General Aptitude: 4 questions, 4 marks
EMFT: 2 questions, 4 marks
Analog Electronics: 3 questions, 3 marks
Digital Electronics: 3 questions, 3 marks
Signals and Systems: 3 questions, 3 marks

Difficulty Distribution:

Easy: 18 | Moderate: 38 | Difficult: 9

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.